



Sharif University of Technology

Department of Electrical Engineering

Course:

ASIC/FPGA

Report:

ASIC/FPGA Project - Group 8

Joint Report with Group 7

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Cross-Group Verification Overview

The final verification phase of the project was a **cross-group verification**: the synthesized OTP Memory Controller was verified against `tb_otp_controller.v`, the independently developed testbench of the cooperating group (Group 7). Rather than re-checking our own environment, this phase evaluated the compatibility and correctness of our implementation against a verification environment written by an outside team with no visibility into our internal design decisions – the strictest functional check the design saw over the course of the project.

Verification was carried out in two stages, both executed on the Cadence Xcelium simulator:

1. **Functional simulation** of the RTL implementation.
2. **Post-synthesis simulation** of the gate-level netlist, with SDF back-annotation.

The cooperating group's testbench exercised six categories: reset behaviour, field extraction, patch override, multiple patches, invalid patches, and CRC error handling.

Initial Cross-Group Results

Functional Simulation (Pre-Synthesis)

Running the design against the cooperating testbench for the first time surfaced a single failure, isolated entirely to the Reset Test:

Test	Errors
Reset Test	2
Field Extraction Test	0
Patch Override Test	0
Multiple Patches Test	0
Invalid Patch Test	0
CRC Error Test	0

Table 1: Cross-group functional simulation results before the fix. Every category except Reset Test passed cleanly on the first run.

```
ERROR: incorrect OTP address sequence: read=312 expected=255 got=254
total cycles =          314
ERROR: otp_read_en pulsed 313 times, expected 312
(256 bank + 56 patch)
Reset Test completed with 2 errors!
```

Failure Signature The cooperating testbench expected exactly **312 OTP read operations** (256 bank bits + 56 patch bits). Our controller instead generated **313 otp_read_en pulses** over **314 cycles** – one pulse and one cycle too many – and the sequence checker flagged that address **254** was issued twice at the end of the read sequence.

Post-Synthesis Simulation

The same suite was then re-run against the synthesized gate-level netlist with SDF back-annotation. The result was an exact reproduction of the RTL failure – same error text, same cycle count, same pulse count:

```
ERROR: incorrect OTP address sequence: read=312 expected=255 got=254
total cycles = 314
ERROR: otp_read_en pulsed 313 times, expected 312
(256 bank + 56 patch)
Reset Test completed with 2 errors!
```

All other tests again completed with zero errors. Because the discrepancy was **bit-for-bit identical** between RTL and gate-level simulation, this immediately ruled out synthesis, timing, or SDF back-annotation as the cause: the bug had to be a functional issue already present in the RTL, not something introduced by the synthesis flow.

Root Cause Analysis

The two reported errors – the repeated address and the extra read-enable pulse – were traced to a **single underlying FSM timing issue**, not two independent bugs.

The controller carries a pair of delayed signals, `addr_d` and `ren_d`, whose purpose is to align accumulated data with the simulated OTP memory’s timing: the simulation ROM registers `otp_data_out` one clock cycle *after* `otp_addr` and `otp_read_en` are presented, so the one-cycle-delayed copies are needed for correct data accumulation.

The bug was that the FSM’s *next-state* logic also used these delayed signals to decide when the read states should terminate, while the address counter itself advanced on the live, non-delayed `addr` signal:

Listing 1: FSM Next-State Logic Before the Fix – Uses Delayed Signals for Read Termination

```

1 STATE_READ_BANK0:
2     next_state = (ren_d && addr_d == 8'd255) ?
3                 STATE_READ_PATCH : STATE_READ_BANK0;
4
5 STATE_READ_PATCH:
6     next_state = (ren_d && addr_d == PATCH1_HI) ?
7                 STATE_APPLY : STATE_READ_PATCH;
```

Causal Chain:

1. The live address counter reached the final patch address.
2. The live address counter stopped advancing, having hit its terminal value.
3. The FSM, however, was still waiting for the *delayed* address signal to reach that same terminal value – one cycle behind.
4. Consequently, the final address was re-presented for one additional cycle while `otp_read_en` remained asserted.
5. The cooperating testbench’s strict sequence checker flagged the repeated address, and its pulse counter flagged the extra `otp_read_en` assertion.

One Root Cause, Two Symptoms The repeated address and the extra read pulse are two observations of the *same* off-by-one condition: the FSM’s termination check ran one cycle later than the address counter’s own termination, because it was keyed to the delayed alignment signals instead of the live ones.

Impact of the Bug

The issue was confined to the **control-flow timing** of the OTP read sequence and did not corrupt any accumulated data: the final patch bit was sampled and accumulated twice, but with the identical value both times, so downstream processing was unaffected.

- **Timing only:** one additional clock cycle and one additional `otp_read_en` pulse in the read sequence, with the final OTP address repeated.
- **No data corruption:** field extraction, patch processing, CRC handling, and patch override behaviour were all unaffected – every non-Reset test passed both before and after the fix.
- **Compliance, not correctness:** the failure was a violation of the cooperating test-bench's strict sequence and cycle-count checks, not evidence of incorrect functional behaviour.

The bug was therefore classified as a **control-sequencing and verification-compliance bug** rather than a data-integrity bug.

RTL Correction and Re-Verification

Fix Applied

The correction made the FSM's termination condition consistent with the address counter it was meant to track: the state-transition logic was switched from the delayed alignment signals to the same live signals already driving the address counter, while `addr_d` and `ren_d` were left untouched for their original data-accumulation role.

Listing 2: FSM Next-State Logic After the Fix – Uses Live Signals for Read Termination

```

1 STATE_READ_BANK0:
2     next_state = (otp_read_en && addr == 8'd255) ?
3                 STATE_READ_PATCH : STATE_READ_BANK0;
4
5 STATE_READ_PATCH:
6     next_state = (otp_read_en && addr == PATCH1_HI) ?
7                 STATE_APPLY : STATE_READ_PATCH;

```

Verification After the Fix

Re-running the cooperating testbench against the corrected RTL cleared the Reset Test entirely:

```

1 ===== Reset Test =====
2 Read started: addr=0, read_en=1
3 total cycles =          313
4 Reset Test completed with          0 errors!
5 ===== Reset Test Complete =====

```

Test	Errors
Reset Test	0
Field Extraction Test	0
Patch Override Test	0
Multiple Patches Test	0
Invalid Patch Test	0
CRC Error Test	0

Table 2: Cross-group verification results after the RTL correction – the complete cooperating test suite passes with zero errors.

Before/After Comparison

Metric	Before Fix	After Fix
Expected OTP read pulses	312	312
Actual OTP read pulses	313	312
Reset Test errors	2	0
Repeated final address	Yes	No
Reset Test cycles	314	313
Field Extraction Test	Pass	Pass
Patch Override Test	Pass	Pass
Multiple Patches Test	Pass	Pass
Invalid Patch Test	Pass	Pass
CRC Error Test	Pass	Pass

Table 3: Cross-group verification outcome before and after the RTL correction.

The post-fix results confirm the FSM now transitions at the correct cycle, eliminating both the extra read pulse and the repeated final address, with every other test category unaffected by the change.

Conclusion

Cross-group verification against Group 7's independently developed testbench successfully surfaced a subtle FSM timing inconsistency that our own regression had not exercised: the OTP controller passed every test related to field extraction, patch handling, and CRC error processing, but the cooperating testbench's stricter sequence and cycle-count checks caught an extra read cycle during reset.

The root cause was a mismatch between the FSM's use of delayed alignment signals (`addr_d`, `ren_d`) for read-termination decisions and the address counter's use of the corresponding live signals, causing the FSM to linger for one extra cycle and re-issue the final address. The fix re-keyed the FSM's termination logic to the live `addr` and `otp_read_en` signals, leaving the delayed signals in place for their original data-accumulation purpose.

After the correction, the Reset Test completed with zero errors and every other test in the cooperating suite continued to pass, confirming that the corrected OTP Memory Controller fully satisfies the cross-group verification requirements and that the previously observed issue has been resolved.